EXPERIMENT 12 - PYTHON CODE

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# EXPERIMENT 12 - Pattern Matching in a Healthcare Symptom Assessment System
import re

symptom_patterns = {
    "Emergency Warning Signs": [
        r"\bchest pain\b", r"\bdifficulty breathing\b",
        r"\bshortness of breath\b", r"\bface drooping\b",
        r"\bsudden weakness\b", r"\bsevere bleeding\b"
    ],
    "Respiratory Symptoms": [
        r"\bcough\b", r"\bsore throat\b", r"\brunny nose\b", r"\bfever\b"
    ],
    "Digestive Symptoms": [
        r"\bnausea\b", r"\bvomiting\b", r"\bstomach pain\b",
        r"\babdominal pain\b", r"\bdiarrhea\b"
    ],
    "Neurological or General Symptoms": [
        r"\bheadache\b", r"\bdizziness\b", r"\bfatigue\b", r"\bweakness\b"
    ]
}

guidance = {
    "Emergency Warning Signs": "Seek emergency medical care immediately. Do not delay care while using this sy
    "Respiratory Symptoms": "Consider professional medical advice if symptoms persist, worsen, or are accompan
    "Digestive Symptoms": "Maintain hydration if appropriate and seek medical advice for persistent or worseni
    "Neurological or General Symptoms": "Consider medical assessment if symptoms persist or worsen. Sudden wea
}

def match_symptoms(patient_input):
    text = patient_input.lower()
    categories, symptoms = [], []
    for category, patterns in symptom_patterns.items():
        found = []
        for pattern in patterns:
            match = re.search(pattern, text)
            if match:
                found.append(match.group())
        if found:
            categories.append(category)
            symptoms.extend(found)
    return categories, symptoms

def healthcare_assistant():
    print("HEALTHCARE SYMPTOM ASSESSMENT SYSTEM")
    print("-" * 40)
    print("Describe the symptoms in one sentence.")
    print("Example: I have a fever, cough and sore throat.")
    patient_input = input("\nPatient: ").strip()

    if not patient_input:
        print("System: Please enter a symptom description.")
        return

    print("\nStep 1: Read Patient Description")
    print("  Input:", patient_input)
    categories, symptoms = match_symptoms(patient_input)

    print("\nStep 2: Match Symptom Patterns")
    if not categories:
        print("  Matched symptoms: None")
        print("  System: No predefined pattern was matched.")
        print("  Please describe the symptoms more clearly or consult a healthcare professional.")
        return

    for category in categories:
        print("  Matched category:", category)

    print("\nStep 3: Assessment Report")
    print("  Matched symptoms:", ", ".join(sorted(set(symptoms))))
    print("  Matched categories:", ", ".join(categories))
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    print("\nStep 4: Provide Guidance")
    if "Emergency Warning Signs" in categories:
        print("\nPriority: EMERGENCY WARNING SIGN")
        print("Guidance:", guidance["Emergency Warning Signs"])
    else:
        print("\nPriority: NON-EMERGENCY PATTERN MATCH")
        for category in categories:
            print(category + ":", guidance[category])

    print("\nNote: This system does not diagnose diseases or replace professional medical assessment.")

healthcare_assistant()
```